

Location Privacy Protection Game Against Adversary Through Multi-User Cooperative Obfuscation

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Abstract—In location-based services (LBSs), it is promising for users to crowdsource and share their Point-of-Interest (PoI) information with each other in a common cache to reduce query frequency and preserve location privacy. Yet most studies on multi-user privacy preservation overlook the opportunity of leveraging their service flexibility. This paper is the first to study multiple users' strategic cooperation against an adversary's optimal inference attack, by leveraging mutual service flexibility. We formulate the multi-user privacy cooperation against the adversary as a max-min adversarial game and solve it in a linear program. Unlike the vast literature, even if a user finds the cached information useful, we prove it beneficial to still query the platform to further confuse the adversary. As the linear program's computational complexity still increases superlinearly with the number of users' possible locations, we propose a binary obfuscation scheme in two opposite spatial directions to achieve guaranteed performance with only constant complexity. Perhaps surprisingly, a user with a greater service flexibility should query with a less obfuscated location to add confusion. Finally, we provide guidance on the optimal query sequence among LBS users. Simulation results show that our crowdsourced privacy protection scheme greatly improves users' privacy as compared with existing approaches.

Index Terms—Decentralized privacy preservation, multi-user cooperative crowdsourcing, location-based services with flexibility, max-min adversarial game theory.

I. INTRODUCTION

LOCATION-BASED services (LBSs) offer mobile users customized service recommendations with an integration of mobile users' geographic locations [2]. To provide useful information about points of interests (PoIs) nearby (e.g., night-clubs and restaurants), the LBS platform responds to users' queries using their current locations. Despite the customized service benefits, the usage of LBSs might leak users' private location information, as query data stored in LBS platforms may be revealed to advertisers or hacked by malicious attackers [3].

To preserve users' location privacy, both centralized and decentralized approaches are proposed and studied [4]. The basic

idea of centralized approaches is to introduce a trusted third party (TTP), which protects users' privacy by operating between users and the LBS platform as the anonymizer (e.g., [5], [6], [7], [8]). Such a TTP collects users' original queries and transmits the processed queries to the LBS platform after applying privacy-preserving techniques (e.g., mix zone, pseudonym). Thus the LBS platform cannot identify the users' real locations. However, a single-point failure at the TTP may lead to full privacy leakage of a large group of users [9]. A recent example is the leakage of user credentials from Okta, a third-party company that handles log-ins for more than 100 million users [10].

Decentralized approaches (e.g., [11], [12], [13]) no longer rely on a TTP between the mobile users and the LBS platform. There, a distributed user needs to initiate the LBS query by applying privacy-preserving techniques himself (e.g., cloaking [14], dummy generation [15], k -anonymity [16] [17] and differential privacy [18][19]). However, such approaches may incur high overhead on the individual. For example, the k -anonymity technique expects a user to obfuscate his real location with $k - 1$ dummy locations, leading to a high complexity for computation and implementation at the end device.

Recently, some simple kinds of decentralized privacy-preserving approaches are proposed by using caching (e.g., [20], [21], [22], [23], [24], [25]). In [20], a user caches his prior query data to answer similar queries in the future and reduce the chances to leak his privacy to the untrusted or compromised LBS platform. In practice, however, an individual user's caching is far from enough to cover many PoIs for him to visit later. Thanks to crowdsourcing, it is more efficient for many users to share their queried PoIs with each other in a common cache. Shokri et al. in [21] proposed a user-collaborative privacy-preserving approach: once a user finds the formerly cached PoI information by other users helpful, he will no longer query the LBS platform. Only if the user finds the shared PoI information in the cache not useful, he has to query the LBS platform using his real location, which can be overheard by the adversary. Provided with users' overlapped mobility patterns and similar LBS interests, such cooperative approach efficiently reduces the overall query frequency for all users. If we relax to allow a user to make multiple queries and bear extra computational complexity or communication overhead, there are some other fusion works further combining caching with k -anonymity [22], [23], [24] or l -diversity [25].

It should be noted that once a user finds useful information in the cache, all the existing caching-based privacy protection approaches ([20], [21], [22], [23], [24], [25]) simply ask him to

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hide from the LBS platform without any query. We wonder if such hiding is beneficial to users, and this motivates the first key question of this paper.

Question 1. If a user already finds the shared PoI information useful in the cache, is it beneficial for him to hide from or further query the LBS platform?

Actually, hiding from the LBS platform without query also reveals that the user's location is already covered in the existing cache, and we will prove in the paper that always querying helps confuse the adversary's optimal inference attack.

On the other hand, if the user does not find useful PoI information in the cache, we wonder how to add obfuscation to his query to still protect his privacy.

Question 2. If a user finds the shared PoI information not useful in the cache, how to strategically add more obfuscation to his query?

In the real world, many LBS users are actually flexible in service requirements and only expect that the returned PoIs are within a certain distance (e.g., restaurants and hotels within 1 km) [26], [27]. An LBS user may query with an obfuscated location instead of his real location to leverage his service flexibility [28]. Some work further models this flexibility as a service requirement constraint [29]. Yet most studies on multi-user privacy preservation overlook the opportunity of leveraging service flexibility. To our best knowledge, this paper is the first to leverage service flexibility in multi-user privacy protection.

On the other hand, the LBS platform or any other cooperative peers may be compromised by the adversary to leak the shared information in the cache and should be regarded as untrusted. The adversary is also aware of the users' service flexibility to add obfuscation to their queries, and may adaptively change the inference attack strategy. To proactively design the multi-user privacy protection mechanism, we should be first prepared to understand the adversary's best inference attack. This leads to the third key question of this paper:

Question 3. What is the adversary's best inference attack to the multi-user strategic LBS querying?

It is natural to use game theory to model the interaction between the users' strategic queries and the adversary's inference attack. The adversary might hack into other untrusted peers or the platform for reported locations in the cache, yet it does not know users' private locations. Then we will model the interaction between users and the adversary as a max-min adversarial Bayesian game and accordingly design users' robust query strategies.

The key novelty and the main results of the paper are summarized as follows.

- *Multi-user privacy cooperation by Leveraging service flexibility:* To the best of our knowledge, this is the first paper to study how multiple users cooperate to query with maximum obfuscation against the adversary's optimal inference attack, by leveraging their mutual service flexibility. We instruct users to not only share their searched PoI information in a common cache to reduce overall query frequency, but also cooperate to add maximum obfuscation to their queries in LBS. We consider the robust defence against an

intelligent adversary, who knows the cached data by former users and the users' objective functions to reverse-engineer and infer users' locations from their queries.

- *Adversarial Bayesian game against optimal inference attack:* As the users' locations are private information to be inferred by the adversary, we naturally formulate multi-user privacy cooperation against the adversary as a max-min adversarial Bayesian game. We manage to simplify it to a linear programming (LP) problem, yet its computational complexity still increases superlinearly with the number of users' possible locations. We prove it beneficial for users to always query the LBS platform to add maximum obfuscation, even if they already find useful PoI information in the crowdsourced cache.
- *Approximate obfuscation cooperation schemes with low complexity:* To greatly save the complexity and derive the closed-form defence solution, we propose a binary approximate obfuscation scheme with only constant complexity for users located on a one-dimensional (1D) line interval (e.g., avenue or road). Depending on whether a user finds the shared PoI information useful or not, this approximation scheme tells how to misreport his query randomly in two opposite spatial directions. This scheme is easy to implement and we also extend it to users located in the two-dimensional (2D) plane to apply randomized misreporting in four spatial directions.
- *Guaranteed multi-user privacy gain:* Our binary approximate obfuscation scheme is proved to guarantee at least 3/5 of the optimal privacy gain. Perhaps surprisingly, we choose to instruct a user with a greater service flexibility to query with a less obfuscated location to strategically confuse the adversary. We also prove the asymptotic optimum of our approximate obfuscation scheme, as long as there are enough number of crowdsourcing users. Extensive simulations show our scheme significantly outperforms the state-of-the-art schemes.
- *Guidelines for the multi-user query sequence:* Besides guiding each user's LBS query location, we further enhance the multi-user privacy protection performance, by optimizing the query sequence of users with different service flexibilities. For the case of two users of similar small service flexibilities to cooperate, we prove it beneficial for the user with less service flexibility to query and help preserve the other user's privacy. Yet the sequence should reverse if they have very diverse service flexibilities. We also simulate the more general multi-user case to show similar insights for the optimal query sequence.

The outline of the paper is organized as follows. Section II presents the system model under the multi-user privacy preservation. Section III formulates each user's privacy cooperation problem against the adversary. Section IV studies a approximate obfuscated query scheme in closed-form in a one-dimensional line. Section V evaluates the performance of the approximate scheme. Section VI studies the optimal query sequence to maximize the total expected privacy gain. Section VII extends the approximate scheme to the two-dimensional scenario. Finally, Section VIII concludes this paper. Due to space limit, we put

the detailed proofs in the supplementary document, available online.

II. SYSTEM MODEL AND PROBLEM FORMULATION

We consider N active LBS users in a set $\mathcal{N} = \{1, 2, \dots, N\}$ with real locations $x_1, x_2, \dots, x_N$ known to themselves only. They need nearby PoI information in a continuous bounded location set $\mathcal{M}$, which can be either a one-dimensional (1D) avenue line or a two-dimensional (2D) ground plane. To learn useful PoI information from each other, a crowdsourced cache is used to enable a user to post and share his searched PoI information with latter users with similar PoI interests. To best use the cache, it is the best for users to sequentially demand PoI information from the LBS such that the latter can learn from former users' queries.

Next, we first introduce the multi-user crowdsourced query scheme with obfuscation, then introduce users' randomized strategies against the adversary's inference attack in this scheme. Finally, we model the strategic interaction between cooperative users and the adversary as an adversarial Bayesian game.

A. Multi-User Crowdsourced Query Scheme With Obfuscation

Without loss of generality, we suppose user i is the i -th among N users to query. Later in Section VI, besides studying each user's query strategy, we will further study the optimal query sequence among N users from the perspective of the crowdsourcing system to maximize the users' total expected privacy.

Each user $i \in \mathcal{U}$ with real location $x_i \in \mathcal{M}$ queries the LBS platform for nearby PoI information (e.g., hotels), by providing a location $x'_i \in \mathcal{M}$. The LBS platform will return PoI information $I(x'_i)$ at x'_i to the user. In practice, the user is flexible to demand PoIs, and finds PoIs useful as long as they are within a certain distance Q_i from his real location x_i (e.g., [26], [30]). Formally, we give the definition of a user's service flexibility in the following.

Definition 1 (User i 's service flexibility). When searching for PoIs in LBSs, user i with real location x_i is flexible to accept the returned PoI information $I(x'_i)$ at x'_i if $D(x_i, x'_i) \leq Q_i$, where $D(\cdot, \cdot)$ measures the Euclidean distance between any two points, the parameter Q_i measures the user's service flexibility.

One can imagine the scenario when an LBS user searches for nearby restaurant recommendations, usually he does not require the returned PoI information to be exactly at his real location, given he is flexible to walk or drive a certain distance. The user requirements of many LBS applications are only needed to be satisfied by different levels of accuracy [27]. This allows a user to misreport a different location x'_i with $x'_i \neq x_i$ satisfying $D(x_i, x'_i) \leq Q_i$ for privacy concerns, bringing in degraded but acceptable service quality. Such location perturbation or obfuscation is one of the common practices in location privacy protection mechanisms [31]. To meet users' different service flexibilities, popular LBS apps such as Yelp also provides choices of different search ranges (0.5 km, 2 km, 5 km, etc.) from a user's entered location to search for nearby PoIs [32].

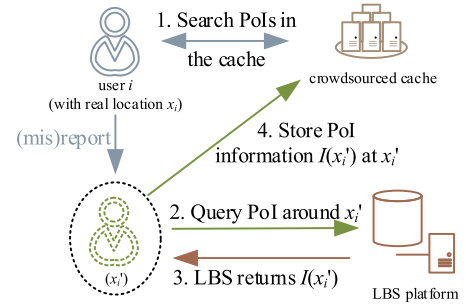


Fig. 1. Users' crowdsourced query scheme in LBS: user i with real location x_i examines the crowdsourced cache first, finding useful/unuseful PoI information. Then he may query the LBS platform with an obfuscated location x'_i and receive the PoI information $I(x'_i)$ at location x'_i . To benefit latter users, he stores $(x'_i, I(x'_i))$ to share in the cache.

Provided with the crowdsourced cache, each user has access to all former users' searched PoI information. Then user i benefits from the multi-user cooperation as long as he finds any former user j 's query x'_j within distance Q_i , i.e., $\exists j = 1, \dots, i-1$ such that

$$D(x_i, x'_j) \leq Q_i. \quad (1)$$

Fig. 1 illustrates our crowdsourced query scheme works in the following four steps.

- Step 1: User $i \in \mathcal{U}$ first examines the data cached by former users for useful PoI information (if any) within distance Q_i from his real location x_i .
- Step 2: User i may continue to query the LBS platform with a location x'_i , which may be different from x_i . Unlike the literature [20], [21], [22], [23], [24], [25], we allow the user to still query even if he finds the shared PoIs useful. Actually, hiding without any query x'_i is a special case of our query strategy here. The user may generally use different query strategies, depending on whether he finds cached PoIs useful or not.
- Step 3: The platform returns PoI information $I(x'_i)$ to the user upon request.
- Step 4: User i stores the query location x'_i and the corresponding PoI information $I(x'_i)$ in the cache to benefit latter users.

After observing $i-1$ former users' shared PoI information in the crowdsourced cache and querying the LBS platform at x'_i , user i is believed to receive an expected privacy gain π_i . Next we will introduce any user i 's randomized strategy against the optimal inference attack to model the expected privacy gain π_i at the i -th order.

B. Users' Randomized Strategies Against the Adversary

Given former users' query summary $\mathbf{x}'_{i-1} = (x'_1, \dots, x'_{i-1})$, it is user i 's turn to decide how to query. Depending on whether $\mathbf{x}'_{i-1}$ includes useful PoI information within walking distance Q_i from x_i , user i generally take two different query strategies in Step 2 above. Mathematically, we define

$$\mathcal{X}_i^{in}(\mathbf{x}'_{i-1}, Q_i) = \cup_{j=1}^{i-1} \{x \in \mathcal{M} | D(x, x'_j) \leq Q_i\}, \quad (2)$$

which summarizes the covered location set from earlier queries for user i 's real location x_i . If $x_i \in \mathcal{X}_i^{\text{in}}(\mathbf{x}'_{i-1}, Q_i)$, user i already finds the cached PoI information useful and meet his service constraint. Similarly, we define

$$\mathcal{X}_i^{\text{out}}(\mathbf{x}'_{i-1}, Q_i) = \mathcal{M} \setminus \mathcal{X}_i^{\text{in}}(\mathbf{x}'_{i-1}, Q_i)$$

as the uncovered PoI location set for user i . If $x_i \in \mathcal{X}_i^{\text{out}}(\mathbf{x}'_{i-1}, Q_i)$, user i has to query with a new x'_i nearby to meet his service constraint $D(x_i, x'_i) \leq Q_i$. Note that the first-query user 1 finds the cache empty with $\mathcal{X}_1^{\text{in}} = \emptyset$, $\mathcal{X}_1^{\text{out}} = \mathcal{M}$.

To determine the query location x'_i for any given x_i , we generally use a conditional probabilistic distribution $f_i(x'_i|x_i)$ to denote user i 's randomized query strategy. It is a mapping probability from $x_i \in \mathcal{M}$ to $x'_i \in \mathcal{M}$, depending on $x_i \in \mathcal{X}_i^{\text{in}}(\mathbf{x}'_{i-1}, Q_i)$ or $x_i \in \mathcal{X}_i^{\text{out}}(\mathbf{x}'_{i-1}, Q_i)$. It should be noted that in general no user will use a one-to-one deterministic query strategy as the adversary can easily infer his actual location x_i from reported x'_i .

Definition 2 (User i 's randomized query strategy). Depending on whether the cached PoIs are useful, we generally define user i 's query strategy to the LBS platform as

$$\begin{aligned} & f_i(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) \\ &= \begin{cases} f_i^{\text{in}}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i), & \text{if } x_i \in \mathcal{X}_i^{\text{in}}(\mathbf{x}'_{i-1}, Q_i), \\ f_i^{\text{out}}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i), & \text{if } x_i \in \mathcal{X}_i^{\text{out}}(\mathbf{x}'_{i-1}, Q_i). \end{cases} \end{aligned} \quad (3)$$

after observing $I(\mathbf{x}'_{i-1})$ shared by all $i-1$ former users.

As a special case of Definition 2, if the user hides from the LBS platform without any query as in [20], [21], [22], [23], [24], [25], $f_i^{\text{in}}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) = \emptyset$ yet the adversary still learns $x_i \in \mathcal{X}_i^{\text{in}}(\mathbf{x}'_{i-1}, Q_i)$.

Provided with the randomized query strategy in Definition 2, user i 's expected privacy gain also depends on the adversary's optimal inference attack. The adversary can access all former users' queries $\mathbf{x}'_{i-1}$ as well as user i 's query x'_i . Let $\hat{x}_i$ denote the adversary's optimal inference, which is a function of both x'_i and $\mathbf{x}'_{i-1}$. To minimize the inference error, i.e., the expectation of random distance $D(\hat{x}_i, x_i)$ from the inferred location $\hat{x}_i$ to user i 's real location x_i , the adversary's optimal inference problem for user i is given as

$$\min_{\hat{x}_i \in \mathcal{M}} \int_{x_i \in \mathcal{M}} \Pr(x_i|x'_i, \mathbf{x}'_{i-1}) D(\hat{x}_i, x_i) dx_i, \quad (4)$$

where $\Pr(x_i|x'_i, \mathbf{x}'_{i-1})$ is a posterior probability of user i 's real location x_i . We will analyze this attack in detail later in Section III-A.

Let $\psi_i(x_i)$ denote the probability density function (PDF) of user i 's random location x_i , which is also known to the adversary by checking historical query data. Then user i 's expected privacy gain π_i is defined as the expectation of random distance $D(x_i, \hat{x}_i(x'_i|\mathbf{x}'_{i-1}))$ from his real location x_i to the adversary's optimal inference $\hat{x}_i$, by averaging over all possible x_i and x'_i by using two query strategies in Definition 2. That is,

$$\begin{aligned} \pi_i &= \int_{x'_i \in \mathcal{M}} \int_{x_i \in \mathcal{X}_i^{\text{in}}(\mathbf{x}'_{i-1}, Q_i)} \psi_i(x_i) f_i^{\text{in}}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) \\ &\quad D(x_i, \hat{x}_i(x'_i|\mathbf{x}'_{i-1})) dx_i dx'_i \\ &\quad + \int_{x'_i \in \mathcal{M}} \int_{x_i \in \mathcal{X}_i^{\text{out}}(\mathbf{x}'_{i-1}, Q_i)} \psi_i(x_i) f_i^{\text{out}}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) \\ &\quad D(x_i, \hat{x}_i(x'_i|\mathbf{x}'_{i-1})) dx_i dx'_i \\ &= \int_{x'_i \in \mathcal{M}} \int_{x_i \in \mathcal{M}} \Pr(x_i, x'_i|\mathbf{x}'_{i-1}) \\ &\quad D(x_i, \hat{x}_i(x'_i|\mathbf{x}'_{i-1})) dx_i dx'_i. \end{aligned} \quad (5)$$

The last equality is due to the product rule on conditional probability: $\Pr(x_i, x'_i) = \psi_i(x_i) f_i(x'_i|x_i)$. Next we are ready to model the strategic interaction between N cooperative users and the adversary as an adversarial Bayesian game.

C. Adversarial Bayesian Game Formulation

Based on the adversary's optimal inference attack formulation in (4), we formally model the strategic interaction between any user i and the adversary as an adversarial Bayesian game with two stages:

- In Stage I, user i decides a probabilistic strategy $f_i(x'_i|x_i, \mathbf{x}'_{i-1}) = \{f_i^{\text{in}}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i), f_i^{\text{out}}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)\}$ in (3) to query the LBS platform. The objective is to maximize his expected privacy gain in (5), depending on whether he finds the cached PoI information useful or not.
- In Stage II, without the knowledge of user i 's real location x_i but prior distribution $\psi_i(x_i)$, the adversary launches its optimal Bayesian inference attack to infer user i 's location as $\hat{x}_i$. To provide robust privacy preservation ([33], [34]), we look at the challenging case that the adversary has access to the cached data $\mathbf{x}'_{i-1}$ upon user i 's query, and knows users' privacy gain function in (5). After observing user i 's query location x'_i as well as former users' queries $\mathbf{x}'_{i-1}$, its objective is to minimize its inference error in (4).

In our multi-user crowdsourced query scheme, there is no conflict between any two users, as each user sequentially demands PoI and maximizes the individual privacy gain.

So far we have modelled the strategic interaction between cooperative users and the adversary as a two-stage Bayesian game, where users move first by making the query against the adversary's inference attack. Next in Section III, we will analyze the Bayesian game by backward induction.

III. MULTI-USER COOPERATION AGAINST ADVERSARIAL INFERENCE ATTACK

In this section, we first analyze the adversary's optimal inference attack $\hat{x}_i$. Then we proactively design each user i 's two different query strategies f_i^{in} and f_i^{out} in (3), by taking the adversary's optimal attack response $\hat{x}_i$ into account. Finally, we discuss on whether to hide from the LBS platform or not if the user already finds useful information in the cache.

A. Adversary's Optimal Bayesian Inference Attack

To ensure reliable performance for the user, we consider the worst-case of the fully informed adversary, which is a standard

approach of modelling robust privacy defence [33], [34]. By accessing the cache, any cooperative peers and/or the historical data from the LBS platform, the adversary can easily have full knowledge of the reported data $\mathbf{x}'_i$, each user's privacy gain function as in (5), all users' privacy objective functions and their location distributions $\psi_i(x_i)$ ($i = 1, \dots, N$). Thus it can estimate users' query strategies $f_i(x'_i|x_i, \mathbf{x}'_{i-1})$ on their behalves, and launch the optimal Bayesian inference attack of each user's real location.

After observing user i 's query location x'_i as well as $i-1$ former users' queries $\mathbf{x}'_{i-1}$, the adversary updates the posterior probability of user i 's real location x_i below:

$$\begin{aligned} \Pr(x_i|x'_i, \mathbf{x}'_{i-1}) &= \frac{\Pr(x_i, x'_i|\mathbf{x}'_{i-1})}{\Pr(x'_i|\mathbf{x}'_{i-1})} \\ &= \frac{\psi_i(x_i)f_i(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)}{\int_{x_i \in \mathcal{M}} \psi_i(x_i)f_i(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)dx_i}, \end{aligned} \quad (6)$$

where the denominator is the integral over the whole location region $\mathcal{M}$. The strategy f_i equals $f_i^{in}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)$ for $x_i \in \mathcal{X}_i^{in}(\mathbf{x}'_{i-1}, Q_i)$ or $f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)$ for $x_i \in \mathcal{X}_i^{out}(\mathbf{x}'_{i-1}, Q_i)$ in (3). As a special case, for the first user to arrive and query with $\mathcal{X}_1^{in} = \emptyset$ and $\mathcal{X}_1^{out} = \mathcal{M}$, (6) reduces to

$$\Pr(x_1|x'_1) = \frac{\Pr(x_1, x'_1)}{\Pr(x'_1)} = \frac{\psi_1(x_1)f_1^{out}(x'_1|x_1)}{\int_{x_1 \in \mathcal{M}} \psi_1(x_1)f_1^{out}(x'_1|x_1)dx_1}.$$

By substituting the posterior probability in (6) to (4), we can analyze the adversary's optimal inference problem for user i in (4). By taking the adversary's inference into consideration, we next reformulate each user's obfuscated query problem in (5) as a max-min optimization problem.

B. Max-Min Problem Formulation and Simplification

By taking the adversary's inference problem in (4) using (6) into consideration, we rewrite any user i 's expected privacy gain π_i from (5) to:

$$\begin{aligned} \pi_i &= \int_{x'_i \in \mathcal{M}} \Pr(x'_i|\mathbf{x}'_{i-1}) \\ &\quad \min_{\hat{x}_i \in \mathcal{M}} \int_{x_i \in \mathcal{M}} \Pr(x_i|x'_i, \mathbf{x}'_{i-1})D(\hat{x}_i, x_i)dx_i dx'_i \\ &= \int_{x'_i \in \mathcal{M}} \min_{\hat{x}_i \in \mathcal{M}} \int_{x_i \in \mathcal{M}} \psi_i(x_i)f_i(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) \\ &\quad D(\hat{x}_i, x_i)dx_i dx'_i, \end{aligned} \quad (7)$$

which integrates over any possible x'_i . Both equalities in (7) hold due to the product rule on conditional probability: $\Pr(x_i, x'_i) = \Pr(x'_i)\Pr(x_i|x'_i)$ for the first equality and $\Pr(x_i, x'_i) = \psi_i(x_i)f_i(x'_i|x_i)$ for the second equality.

User i aims to optimize its expected privacy gain π_i in (7), by considering his PoI location requirement within distance Q_i from x_i :

$$\max \int_{x'_i \in \mathcal{M}} \min_{\hat{x}_i \in \mathcal{M}} \int_{x_i \in \mathcal{M}} \psi_i(x_i)f_i(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)D(\hat{x}_i, x_i)dx_i dx'_i$$

$$\begin{aligned} \text{s.t. } &f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) = 0, \forall (x_i, x'_i) \in \\ &\{(x_i, x'_i)|x_i \in \mathcal{X}_i^{out}(\mathbf{x}'_{i-1}, Q_i), x'_i \in \mathcal{M}, D(x'_i, x_i) > Q_i\}, \\ &\int_{x'_i \in \mathcal{M}} f_i(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)dx'_i = 1, \forall x_i \in \mathcal{M}, \\ &\text{var} : f_i(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) \text{ in } (3), \forall x_i, x'_i \in \mathcal{M}. \end{aligned} \quad (8)$$

The first constraint of (8) is to meet the service requirement of user i . If he finds his real location $x_i \in \mathcal{X}_i^{in}(\mathbf{x}'_{i-1}, Q_i)$, this requirement is met regardless of his query strategy f_i^{in} . Otherwise, if $x_i \in \mathcal{X}_i^{out}(\mathbf{x}'_{i-1}, Q_i)$, we should ensure that user i will not report a location x'_i with more than distance Q_i away from x_i . Thus, we require $D(x'_i, x_i) \leq Q_i$, or equivalently zero probability $f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) = 0$ for $D(x'_i, x_i) > Q_i$.

As the location region $\mathcal{M}$ is continuous, it is difficult to solve two continuous variable functions $f_i^{in}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)$ and $f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)$ in (8). Instead, we propose the following alternative.

Proposition 1. By equally partitioning the continuous location set $\mathcal{M}$ into M discrete grids, we simplify the objective of Problem (8) as:

$$\begin{aligned} &\max \sum_{x'_i} \min_{\hat{x}_i} \sum_{x_i \in \mathcal{M}} \psi_i(x_i) \\ &f_i(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)D(\hat{x}_i, x_i), \end{aligned} \quad (9)$$

then the problem (8) for each user i becomes a linear program (LP) with computational complexity $\mathcal{O}(M^7)$.

The time complexity of (9) is derived as inspired by [35], which solved an LP problem with n variables and m constraints by an interior algorithm within complexity $\mathcal{O}(m^{3/2}n^2)$. By using some toolboxes (e.g., Optimization Toolbox in MATLAB), we can solve our LP problem numerically and our solution approaches the optimum as M goes to infinity to reduce the discretization error. Despite of the error, this LP may not be solvable for a large-scale spatial region and we will propose an approximate obfuscated query solution on the original problem (8) later in Section IV.

C. To Hide or Not When $x_i \in \mathcal{X}_i^{in}(\mathbf{x}'_{i-1}, Q_i)$

Recall that in the existing multi-user privacy preservation schemes in [20], [21], [22], [23], [24], [25], users query the LBS platform only if the target PoI information is not found in the cache. Their hiding-from-the-LBS idea leads to $f_i^{in} = \emptyset$ in (3) in our problem formulation as there is no query for $x_i \in \mathcal{X}_i^{in}(\mathbf{x}'_{i-1})$. In this case, the adversary can still infer user i 's location in the covered region $\mathcal{X}_i^{in}(\mathbf{x}'_{i-1})$. We can similarly formulate a max-min problem for the hiding scheme as (8), by only limiting the decision variable function to f_i^{out} for the case of $x_i \in \mathcal{X}_i^{out}(\mathbf{x}'_{i-1}, Q_i)$. Let $\mathbb{E}\pi^{opt} = \frac{1}{N} \sum_i \pi_i^{opt}$ denote the average user's privacy gain under the optimal solution to problem (8). Let $\mathbb{E}\tilde{\pi}^{opt}$ denote the counterpart from problem (9) after discretization. Then when M is large enough, $\mathbb{E}\tilde{\pi}^{opt}$ approaches $\mathbb{E}\pi^{opt}$. We also let $\mathbb{E}\pi^{hide}$ denote the average user's privacy gain under the existing hiding schemes as in [20], [21],

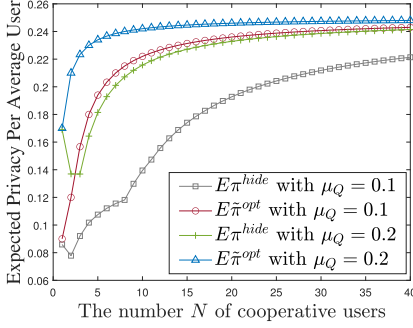


Fig. 2. Performance comparison between our strategy's performance $\mathbb{E}\pi^{opt}$ to Problem (9) and the existing hiding strategy's performance $\mathbb{E}\pi^{hide}$ in the literature ([20], [21], [22], [23], [24], [25]). We set each user i 's service flexibility constraint Q_i to follow the i.i.d. truncated normal distribution with mean $\mu_Q = 0.1$ or 0.2 under the standard deviation $\sigma = 0.1$ among N users. The discretization size for $\mathcal{M}$ is $M = 20$.

[22], [23], [24], [25]. Next we analytically compare the existing hiding solution with to our always-query solution.

Lemma 1. $\mathbb{E}\pi^{opt} > \mathbb{E}\pi^{hide}$ always holds for $N > 1$ and it is always beneficial for each user to strategically query the LBS platform with $f_i^{in} \neq \emptyset$ in (3).

Lemma 1 answered our first key question in the introduction that even if the user already gets help from the crowdsourced cache, querying the LBS platform strategically helps improve the privacy gain further. The intuitions behind are explained in the two following aspects:

- *Adding confusion to the adversary:* When the user always queries, the adversary does not know $x_i \in \mathcal{X}_i^{in}(x'_{i-1}, Q_i)$ or $x_i \in \mathcal{X}_i^{out}(x'_{i-1}, Q_i)$. The two randomized strategies $f_i^{in}(x'_i|x_i, x'_{i-1}, Q_i)$ and $f_i^{out}(x'_i|x_i, x'_{i-1}, Q_i)$ in (3) jointly confuse the adversary.
- *More PoIs to benefit latter users:* Our always-query strategy returns more PoI data in the cache for latter users to take advantage of. Thus the covered PoI location sets are enlarged for latter users to use.

Besides the analytical comparison in Lemma 1, we run simulations in Fig. 2 to empirically compare the tractable performance $\mathbb{E}\pi^{opt}$ by solving Problem (9) with $\mathbb{E}\pi^{hide}$ for any user number N . Here all users' service flexibility constraints Q_i follow the i.i.d. truncated normal distribution with the minimum value $Q_i \geq 0$. Given the same standard deviation, we examine the performances under different mean values μ_Q of Q_i distribution.

Due to the PoI sharing benefit in multi-user cooperation, all privacy performance curves (no matter for hiding strategy or not) are generally increasing in user number N . The expected privacy gain increases in the mean value μ_Q of Q_i , as a greater flexibility helps users add obfuscation to their queries strategically for a better privacy gain. An average user's expected privacy gain $\mathbb{E}\pi^{opt}$ obviously improves under our always-query strategy, as compared to the traditional $\mathbb{E}\pi^{hide}$ with hiding. The performance advantage becomes obvious for a non-small N , as the our always-query strategy creates the maximum obfuscation to the adversary and creates more PoIs to share among users.

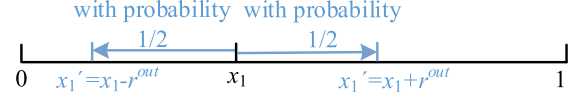


Fig. 3. User 1's query strategy $f_1^{out}(x'_1|x_1, \emptyset)$ with two randomized queries $x'_1 = |x_1 - r^{out}|$ or $x'_1 = 1 - |x_1 - 1 + r^{out}|$ with equal probability $\frac{1}{2}$, in two spatial directions of x_1 .

On the other hand, the hiding strategy from the LBS platform may expose the user's real location to the cache-covered location set $\mathcal{X}_i^{in}$ and facilitate the adversary's inference attack. Note that in Fig. 2 $\mathbb{E}\pi^{hide}$ even decreases from $N = 1$ to 2 for both curves $\mu_Q = 0.1$ and $\mu_Q = 0.2$. It is because that the second user's possible hiding exposes the user's narrowed location region in $\mathcal{X}_2^{in}(x'_1)$ and helps the adversary to locate his infer $\hat{x}_2$ around the former user 1's query x'_1 . Only after more users' joining can the cooperation benefits outweigh the hiding disadvantage.

IV. APPROXIMATE OBFUSCATED QUERY SCHEME

To make the problem (8) solvable for a large-scale continuous region and provide clean engineering insights, this section presents our approximate obfuscated query scheme in closed-form. Without much loss of generality, we first assume the users' location set $\mathcal{M}$ to be a normalized 1D line interval $\mathcal{M} = [0, 1]$. We will similarly extend our analysis and solution to a 2D plane in Section VII later. We also assume that each user is equally located at any point in interval $\mathcal{M}$ by following an i.i.d. uniform distribution with PDF $\psi_i(x_i) = 1$. Similar results and algorithms can be extended to arbitrary location distributions.

Next we will start with analyzing the first user's approximate query strategy facing an empty cache in Section IV-A. Then we will analyze for the second user and latter users in Sections IV-B and IV-C, respectively. Instead of deciding continuous functions f_i^{in} and f_i^{out} in (3) for user $i \in \mathcal{N}$ over the location set $\mathcal{M}$, our key approximation idea is to reduce each of them to be the randomization of only two location points in two opposite spatial directions from x_i .

A. The First User's Random Query Strategy

First, we discuss the approximate query strategy for the first user 1 to demand PoIs, who observes an empty cache (i.e., $\mathcal{X}_1^{in} = \emptyset$) and its strategy only includes f_1^{out} according to Definition 2. To simplify the continuous query strategy $f_1^{out}(x'_1|x_1, \emptyset)$, we approximate user 1's strategy to a two-sided random query scheme (see illustration in Fig. 3) as:

$$f_1^{out}(x'_1|x_1, \emptyset) = \begin{cases} \frac{1}{2}, & \text{if } x'_1 = |x_1 - r^{out}|, \\ \frac{1}{2}, & \text{if } x'_1 = 1 - |x_1 - 1 + r^{out}|, \end{cases} \quad (10)$$

where the absolute term appears to keep query location x'_1 within the interval $\mathcal{M}$. Such binary approximation is the simplest but fundamental way to replace the complicated randomization

function f_1^{out} . To maximizing π_1 in (8), user 1's decision now changes from $f_1^{out}(x'_1|x_1, \emptyset)$ to r^{out} only.

Proposition 2. The optimal obfuscation distance in (10) to maximize the first user's expected privacy gain is

$$r^{out} = \min\left(Q_1, \frac{1}{2}\right). \quad (11)$$

His maximum expected privacy gain is $\pi_1^{appr} = \min(Q_1 - Q_1^2, \frac{1}{4})$.

To add maximum obfuscation, user 1 reports the farthest possible query point x'_1 with maximum distance Q_1 from x_1 , while merely meeting the service constraint. Even knowing r^{out} , the adversary is unsure about x_1 given the binary randomization.

B. The Second User's Random Query Strategy

After observing the historical query data $\mathbf{x}'_{i-1}$ and his covered PoI location set

$$\mathcal{X}_i^{in}(\mathbf{x}'_{i-1}) = (\cup_{j=1}^{i-1} [x'_j - Q_i, x'_j + Q_i]) \cap [0, 1],$$

user $i \in \{2, \dots, N\}$ needs to decide $f_i^{in}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)$ and $f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i)$ in Definition 2. Similar to (10), we apply the following *binary approximation* of users' obfuscated query strategy.

Definition 3 (Binary approximation for obfuscated query). If a user i finds the cached PoI information useful (i.e., $x_i \in \mathcal{X}_i^{in}(\mathbf{x}'_{i-1}, Q_i)$), we approximate his query strategy to randomization of two symmetric points on the two sides of his real location x_i :

$$f_i^{in}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) = \begin{cases} \frac{1}{2}, & \text{if } x'_i = |x_i - r_i^{in}|, \\ \frac{1}{2}, & \text{if } x'_i = 1 - |x_i - 1 + r_i^{in}|. \end{cases} \quad (12)$$

Otherwise, if $x_i \in \mathcal{X}_i^{out}(\mathbf{x}'_{i-1}, Q_i)$,

$$f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1}, Q_i) = \begin{cases} \frac{1}{2}, & \text{if } x'_i = |x_i - r^{out}(Q_i)|, \\ \frac{1}{2}, & \text{if } x'_i = 1 - |x_i - 1 + r^{out}(Q_i)|. \end{cases} \quad (13)$$

Note that this definition also holds for user 1, as he always finds $x_1 \in \mathcal{X}_1^{out}(\mathbf{x}'_0 = \emptyset) = \mathcal{M}$. Such binary approximation is the simplest but fundamental way to replace the complicated randomization functions f_i^{in} and f_i^{out} . For any user i demanding PoIs later than user 1, (e.g., user 2), if $x_i \in \mathcal{X}_i^{in}(\mathbf{x}'_{i-1}, Q_i)$, he should take different strategies of r_i^{in} in (12) according to prior queries $\mathbf{x}'_{i-1}$. If $x_i \in \mathcal{X}_i^{out}(\mathbf{x}'_{i-1}, Q_i)$, as in user 1's case, to add maximum obfuscation, we suppose user i reports the farthest possible query point x'_i with maximum distance Q_i from x_i .

For ease of exposition, we separate the analysis for user 2 and the following users and in the rest of this subsection we focus on user 2 who is the second to query to explain. Next we optimize r_2^{in} for maximizing user 2's expected privacy gain π_2 . To avoid the trivial case that user 2 has sufficient flexibility or large Q_2 in the first constraint of (8) to arbitrarily misreport his location without service loss, we consider a challenging case

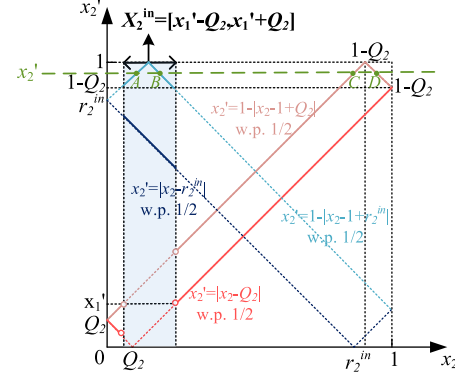


Fig. 4. Example of $Q_2 < x'_1 \leq \frac{1}{2}$ for explaining user 2's approximate query strategy: $f_2^{in}(x'_2|x_2, x'_1)$ with r_2^{in} in (14) in two blue solid lines and $f_2^{out}(x'_2|x_2, x'_1)$ in two red solid lines.

with $Q_2 < 1/11$ here. The analysis can be extended for a larger Q_2 with a better privacy performance.

Proposition 3. Assuming $Q_2 < 1/11$, even if user 2 finds the PoI information shared by user 1 useful (i.e., $D(x'_1, x_2) \leq Q_2$), it is optimal for him to misreport x'_2 with distance $r_2^{in}(x'_1)$ away from x_2 in Definition 3 when querying the LBS platform, where the obfuscation distance is given by

$$r_2^{in}(x'_1) = \begin{cases} 1 - Q_2, & \text{if } 0 \leq x'_1 \leq Q_2, \\ 1 - x'_1, & \text{if } Q_2 < x'_1 \leq \frac{1}{2}, \\ x'_1, & \text{if } \frac{1}{2} < x'_1 \leq 1 - Q_2, \\ 1 - Q_2, & \text{if } 1 - Q_2 < x'_1 \leq 1. \end{cases} \quad (14)$$

As our binary approximation solution is in closed-form, we manage to reduce the computational complexity of deciding each user's query strategy from $\mathcal{O}(M^7)$ in Problem (9) to $\mathcal{O}(1)$ here. Surprisingly, notice that r_2^{in} in (14) is non-increasing in user 2's service flexibility Q_2 . This tells that the user with a greater service flexibility should query the LBS platform with less obfuscated location, for strategically confusing the adversary.

To better explain the implication of r_2^{in} design in (14), we present Fig. 4 to give an example of user 2's query strategy. In this example, user 1's realized query satisfies $Q_2 < x'_1 \leq \frac{1}{2}$, and user 2 has a covered PoI interval $\mathcal{X}_2^{in}(x'_1) = [x'_1 - Q_2, x'_1 + Q_2]$ thanks to user 1's query. If $x_2 \in \mathcal{X}_2^{in}(x'_1)$, user 2's service constraint is met and he will query on the two sides of x_2 randomly with $f_2^{in}(x'_2|x_2, x'_1)$ in (12) with r_2^{in} close to 1 in (14). This strategy $f_2^{in}(x'_2|x_2, x'_1)$ is shown in two blue solid lines (i.e., $x'_2 = |x_2 - r_2^{in}|$ and $x'_2 = 1 - |x_2 - 1 + r_2^{in}|$) which are with equal probability $\frac{1}{2}$. If $x_2 \in \mathcal{X}_2^{out}(x'_1)$, user 2 cannot benefit from the cooperation and he will query with $f_2^{out}(x'_2|x_2, x'_1)$ in (13) with $r^{out}(Q_2) = Q_2$ to satisfy the service requirement. This strategy $f_2^{out}(x'_2|x_2, x'_1)$ is shown in two red solid lines (i.e., $x'_2 = |x_2 - Q_2|$ and $x'_2 = 1 - |x_2 - 1 + Q_2|$) which are with equal probability $\frac{1}{2}$.

In this example, when the adversary observes user 2's query x'_2 as highlighted by the green dash line in Fig. 4, it has four intersections A, B, C, D with user 2's query strategy. This leads to four possible real locations of inferring x_2 with equal probabilities:

the adversary is even not sure whether user 2 is in the covered PoI interval $\mathcal{X}_2^{in}$ (for points A, B) or in $\mathcal{X}_2^{out}$ (for points C, D). Though applying binary approximation, our randomized query strategy makes it difficult for the attacker to launch the optimal inference attack in (4). It is also better than hiding strategy in the literature ([20], [21], [22], [23], [24], [25]), where the adversary immediately infers x_2 in the narrow interval $\mathcal{X}_2^{in}$ (with points A, B) without much obfuscation.

Given the latter user 2's query strategy is adjusted according to user 1's reports x'_1 , the following Corollary 1 tells the impact of x'_1 on user 2's expected privacy gain.

Corollary 1. User 2's expected privacy gain $\pi_2^{appr}(x'_1)$ in the approximate obfuscated query scheme in Definition 3 using r_2^{in} in (14) reaches its minimum when user 1's query point is in the middle of location set $\mathcal{M}$ (i.e., $x'_1 = \frac{1}{2}$).

When x'_1 is at the centre of the location set $\mathcal{M}$, the four possible points for inferring x_2 (see points A, B, C and D in Fig. 4) will be close and user 2 cannot add much obfuscation to its query. This leads to a smaller privacy gain with less confusion to the adversary.

Compared with the existing hiding schemes [20], [21], [22], [23], [24], [25], we can also show the performance of our obfuscation scheme.

Corollary 2. $\pi_2^{appr} > \pi_2^{hide}$ always holds for the second user.

Given the adversary knows the user's full strategy, when observing no query from user 2, it can narrow the location region for user 2 to $\mathcal{X}_2^{in}$ under the hiding-from-the-LBS scheme.

C. Latter Users' Random Query Strategy

For any arbitrary user $i \geq 3$, we similarly apply binary approximation in Definition 3 to simplify the max-min problem (7) and derive his privacy gain π_i^{appr} , by optimizing r_i^{in} in (12). Though more involved, we still follow the backward induction by first analyzing the adversary's optimal inference $\hat{x}_i(x'_i | \mathbf{x}'_{i-1})$ and then maximizing user i 's expected privacy gain π_i^{appr} .

1) Adversary's Optimal Inference Attack Under Definition 3: Similar to Section III-A, the adversary can infer user i 's approximated query strategy in Definition 3 including r_i^{in} in (14) and r_i^{out} in (11). Given the adversary's posterior probability $\Pr(x_i | x'_i, \mathbf{x}'_{i-1})$ of user i 's location x_i in (6), Lemma 2 below summarizes all possible real locations $\hat{\mathcal{X}}_i$ from the perspective of the adversary.

Lemma 2. After observing user i 's query x'_i and former queries $\mathbf{x}'_{i-1}$ in the cache, the adversary believes the location x_i of user $i \geq 3$ is equally likely to appear in the following set:

$$\begin{aligned} \hat{\mathcal{X}}_i(r_i^{in}, r_i^{out}, x'_i, \mathbf{x}'_{i-1}) \\ = \{x_i \in \mathcal{X}_i^{in}(x'_{i-1}, Q_i) | x'_i = |x_i - r_i^{in}| \text{ or } 1 - |x_i - 1 + r_i^{in}|\} \\ \cup \{x_i \in \mathcal{X}_i^{out}(x'_{i-1}, Q_i) | x'_i = |x_i - r_i^{out}| \text{ or } 1 - |x_i - 1 + r_i^{out}|\}. \end{aligned} \quad (15)$$

The adversary's optimal guess of x_i is the mean of these possible locations:

$$\hat{x}_i(r_i^{in}, r_i^{out}, x'_i, \mathbf{x}'_{i-1}) = \text{mean}(\hat{\mathcal{X}}_i(r_i^{in}, r_i^{out}, x'_i, \mathbf{x}'_{i-1})). \quad (16)$$

Algorithm 1: Optimize r_i^{in} for Any User i .

Input: x'_{i-1}, Q_i
Output: r_i^{in}

- 1 Initialization: $r_i^{out} = \min(Q_i, \frac{1}{2})$,
 $\mathcal{X}_i^{in} = \cup_{j=1}^{i-1} [x'_j - Q_i, x'_j + Q_i] \cap [0, 1]$, $\hat{\mathcal{X}}_i = \emptyset$,
- 2 **for** $r_i^{in}=0:\epsilon:1$ **do**
- 3 **for** $x'_i=0:\epsilon:1$ **do**
- 4 Compute $\hat{\mathcal{X}}_i(r_i^{in}, r_i^{out}, x'_i, \mathbf{x}'_{i-1})$ in (15)
- 5 Compute $\hat{x}_i(x'_i) = \text{mean}(\hat{\mathcal{X}}_i)$ in (16)
- 6 Add elements $D(\hat{\mathcal{X}}_i, \hat{x}_i(x'_i))$ to $\tilde{\mathcal{D}}$
- 7 **end**
- 8 $\pi_i^{appr}(r_i^{in}) = \text{mean}(\tilde{\mathcal{D}})$.
- 9 **end**
- 10 $r_i^{in} = \arg \max \pi_i^{appr}(r_i^{in})$

For example, in Fig. 4, after observing x'_2 in the green dash line, the set of user i 's all possible real locations from the perspective of the adversary is $\hat{\mathcal{X}}_2 = \{A, B, C, D\}$. All these points are equally likely to appear given the symmetric query probability in Definition 3.

2) User i 's Expected Privacy Gain Under Definition 3: Given the adversary's optimal inference in (16), we can determine user i 's expected privacy gain in (7) under Definition 3.

Proposition 4. By using the binary approximate query strategy of x'_i in Definition 3, user i 's expected privacy gain in (7) is given by:

$$\begin{aligned} \pi_i^{appr}(r_i^{in}, r_i^{out}, \mathbf{x}'_{i-1}) \\ = \int_{x'_i \in \mathcal{M}} \frac{1}{\int_{x'_i \in \mathcal{M}} |\hat{\mathcal{X}}_i(r_i^{in}, r_i^{out}, x'_i, \mathbf{x}'_{i-1})| dx'_i} \\ \sum_{x_i \in \hat{\mathcal{X}}_i(r_i^{in}, r_i^{out}, x'_i, \mathbf{x}'_{i-1})} D(\hat{x}_i(r_i^{in}, r_i^{out}, x'_i, \mathbf{x}'_{i-1}), x_i) dx'_i. \end{aligned} \quad (17)$$

Algorithm 1¹ returns the optimal obfuscation distance r_i^{in} for Definition 3 with computational complexity $\mathcal{O}(\frac{1}{\epsilon^2})$.

So far, we have finished the optimal design of the binary approximate obfuscated scheme for any user i 's query strategy. Next we move on to the performance evaluation of the scheme returned by Algorithm 1.

V. EVALUATION OF THE APPROXIMATE QUERY

In this section, we first show our approximate scheme guarantees asymptotic optimum, as long as there are sufficient users in cooperation. Then, we compare the approximate solution with the optimal but complicated cooperative strategy to problem (8) in Section III, as well as the existing caching-based schemes in the literature [20], [21], [22], [23], [24], [25].

¹The algorithm can be extended to an arbitrary distribution $\psi_i(x_i)$. The difference is that the adversary will no longer average among all the possible real locations but take a weighted mean based on $\psi_i(x_i)$.

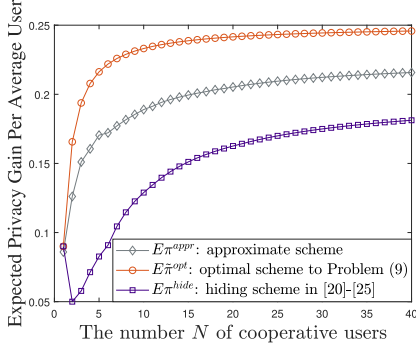


Fig. 5. Performance evaluation of the approximate obfuscated query scheme ($\mathbb{E}\pi^{appr}$) with the optimal strategy ($\mathbb{E}\pi^{opt}$) and hiding schemes in the literature [20], [21], [22], [23], [24], [25]. Users' Q_i follows the i.i.d. truncated normal distribution with mean $\mu = 0.1$ and standard deviation $\sigma = 0.1$.

A. Asymptotic Optimum of Our Approximation Scheme

Lemma 3 shows that a finite number of users are already enough to cover all the PoIs for latter users.

Lemma 3. There exists a finite user number $N' < \infty$ such that for any latter user $i \geq N'$, $\mathcal{X}_i^{in}(\mathbf{x}_{i-1}') = \mathcal{M}$.

As long as we have enough users to cooperate, user $i \geq N'$ at a large enough order can always take advantage of former users' queries to meet the service constraints and hence achieve the maximum possible privacy gain.

Proposition 5. The expected privacy gain for an average user increases with the number N of cooperative users. As $N \rightarrow \infty$, our approximate obfuscated scheme is asymptotically optimal to solve problem (8).

Proposition 5 shows the privacy improvement of our approximate scheme thanks to a great number of users to cooperate and share PoI information.

B. Approximation Ratio Under a Finite User Number

Besides examining the performance for a large user number, we also compare our approximate obfuscated query scheme in Section IV with the optimal solution to Problem (8) in Section III.

As Problem (8) can only be numerically solved without any analytical expression of the optimal performance, we replace with the upper bound of the optimal solution, i.e., $\pi_i^{opt} \leq \frac{1}{4}$ from Proposition 2.

Proposition 6. Our approximate query strategy returned by Algorithm 1 reaches at least $\frac{3}{5}$ of the maximum privacy gain in Problem (8).

Note that this approximation ratio is a loose bound for ease of analysis, and the actual performance ratio is much better.

C. Simulations for Performance Comparison

Fig. 5 presents a numerical example for the expected privacy gain $\mathbb{E}\pi^{appr}$ of the approximate obfuscated query scheme for an average user. As N increases, $\mathbb{E}\pi^{appr}$ approaches the maximum possible privacy gain $\frac{1}{4}$, yet with a diminishing return. It shows that our approximate scheme has limited gap with optimum, yet it saves much time for computing. This gap reduces as we have more users to cooperate, which is consistent with Proposition 5.

Recall in Subsection III-C, we follow the hiding-from-the-LBS idea in the literature to compare with the optimal cooperative strategy. For a fair comparison, here we consider the benchmark case: a user only queries with the binary approximate query strategy when he does not find useful information in the cache and simply hides from the LBS (no query) otherwise as in [20], [21], [22], [23], [24], [25]. We can observe that our approximation scheme obviously outperforms the hiding scheme, to show the advantage of always querying strategically.

VI. OPTIMAL QUERY SEQUENCE AMONG USERS

Besides guiding LBS user's cooperative query strategy, when users come with heterogeneous service flexibility Q_i , we can still optimize the query sequence of users. In this section, we study the optimal query sequence to enhance the multi-user privacy protection performance. Notice that user set $\mathcal{N}$ no longer follows sequence $\{1, \dots, N\}$, and the total expected privacy gain for N users is given as

$$\Pi(\mathcal{N}) = \sum_{i \in \mathcal{N}} \pi_i. \quad (18)$$

First we consider the simple but fundamental scenario with only two users with $Q_1 \leq Q_2$. Recall that in Sections IV-A and IV-B, we analytically solve the approximate query strategy for the users at the first and the second order, and obtain the corresponding privacy gains. To compare two different query sequences to have user 1 or 2 first to query for maximizing (18), we obtain the optimal query sequence.

Proposition 7. When both users have small service flexibilities (i.e., $0 \leq Q_1 \leq Q_2 \leq \frac{1}{11}$), the optimal sequence to maximize the total expected privacy gain is to let user 1 with the smaller service flexibility Q_1 query the LBS platform first.

The intuition behind is that when there are two users with relatively tight service constraints, user 1 with a smaller service flexibility Q_1 can still hardly benefit much from other queries. Yet letting user 1 query first provides a greater service coverage region $\mathcal{X}_2^{in}(\mathbf{x}_1')$ to preserve user 2's privacy.

Proposition 8. When one user has a small service flexibility while the other has a large service flexibility (i.e., $0 \leq Q_1 \leq \frac{1}{11}$ and $Q_2 \rightarrow \frac{1}{2}$), the optimal sequence to maximize the total expected privacy gain is to let user 2 with the greater service flexibility Q_2 query the LBS platform first.

When user 2 have a greater service flexibility Q_2 , his privacy gain already approaches the maximum possible value and can hardly be improved from the cooperation. Thus in this case, we let user 2 query first to benefit latter user 1 with a tighter service constraint. This opposite result from Proposition 7 implies that the optimal query sequence depends on the exact value of users' service flexibilities Q_i and shows no simple monotonicity.

Using the simulation, we can extend such the insights from Propositions 7 and 8 to a more general setting.

Observation 1 (Optimal query sequence for $N = 2$). When $Q_1 = 0.1$ is fixed, there exists two thresholds 0.15 and 0.3 for the value of Q_2 such that

- when $0.15 < Q_2 < 0.3$, the optimal sequence to maximize the total privacy gain of the two users is $\mathcal{N} = \{1, 2\}$, which extends the insight from Proposition 7.

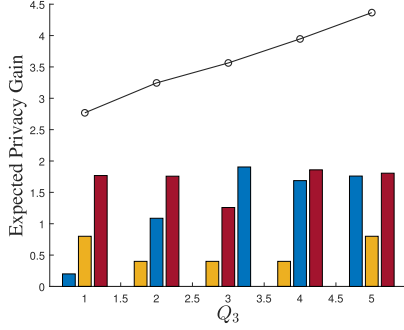


Fig. 6. The optimal sequence for three-user cooperation versus Q_3 when fixing $Q_1 = 0.1$ and $Q_2 = 0.2$.

- when $Q_2 \leq 0.15$ or $Q_2 \geq 0.3$, the optimal sequence to maximize the total privacy gain of the two users is $\mathcal{N} = \{2, 1\}$, which extends the insight from Proposition 8.

To see whether Observation 1 also applies to more than two users, we numerically study on three users' cooperation and obtain the optimal query sequence leading to the maximal total privacy gain Π in Fig. 6. We vary user 3's service flexibility Q_3 while fixing $Q_1 = 0.1$ and $Q_2 = 0.2$. The total expected privacy gain Π increases with Q_3 .

Given users 1 and 2 have similarly small service flexibilities as in Proposition 7, we ask user 1 to query before user 2. If Q_3 is similarly small (i.e., $0 \leq Q_3 < 0.25$), we assign user 3 to be the first, second and third optimally as Q_3 increases (see the blue bar). As Q_3 becomes greater than Q_1 and Q_2 (i.e., $Q_3 > 0.19$), we assign user 3 to be the third, second and first optimally as Q_3 increases. The insight behind is that if user 3's service flexibility Q_3 takes extreme values (i.e., too large or too small), his privacy improvement shows little sensitivity to the query sequence, then he should query the LBS platform first. If Q_3 is relatively in the middle of its value range (i.e., $0.19 < Q_3 < 0.25$), then the user should be the last to query, as his privacy improvement is more sensitive to the query sequence. Similar results can be found for greater user numbers N and the numerical thresholds can be used to guide the crowdsourcing system for a better total privacy performance.

VII. EXTENSION TO 2D LOCATION SCENARIO

Recall in Section IV, we give a binary approximate obfuscated query scheme to design the user's query strategy in a 1D line interval $\mathcal{M} = [0, 1]$ with only left- and right-hand side directions. In this section, we extend the approximate query strategy to the normalized 2D ground plane (i.e., $\mathcal{M} = [0, 1] \times [0, 1]$) using similar design and analysis. For user i with real location (x_i, y_i) , we extend the approximate obfuscated query in four different directions (north, south, east and west) with equal obfuscation distances.

Definition 4 (Four-point approximate obfuscation for LBS query). In the 2D domain, if a user i finds the cached PoI information useful, i.e.,

$$\begin{aligned} & (x_i, y_i) \in \mathcal{X}_i^{in}(\mathbf{x}'_{i-1}, \mathbf{y}'_{i-1}) \\ & = \bigcup_{j=1}^{i-1} \{(x_i, y_i) \in \mathcal{M} | D((x_i, y_i), (x'_j, y'_j)) \leq Q_i\}, \end{aligned}$$

Algorithm 2: Optimize r_i^{in} Under the Four-Point Approximation for Any User i in 2D.

Input: $(x'_{i-1}, y'_{i-1}), Q_i$
Output: r_i^{in}

- 1 Initialization: $r_i^{out} = \min(Q_i, \frac{1}{2})$, $\mathcal{M} = [0, 1] \times [0, 1]$, $\mathcal{X}_i^{in} = \bigcup_{j=1}^{i-1} \{(x_i, y_i) \in \mathcal{M} | D((x_i, y_i), (x'_j, y'_j)) \leq Q_i\} \cap \mathcal{M}$, $\hat{\mathcal{X}}_i = \emptyset$
- 2 **for** $r_i^{in}=0:\epsilon:1$ **do**
- 3 **for** $x'_i=0:\epsilon:1$ **do**
- 4 Compute $\hat{\mathcal{X}}_i(x'_i)$ in (15)
- 5 Compute $\hat{x}_i(x'_i) = \text{mean}(\hat{\mathcal{X}}_i(x'_i))$ in (16)
- 6 **for** $y'_i=0:\epsilon:1$ **do**
- 7 Compute $\hat{\mathcal{Y}}_i(y'_i)$ in (15)
- 8 Compute $\hat{y}_i(y'_i) = \text{mean}(\hat{\mathcal{Y}}_i(y'_i))$ in (16)
- 9 Add elements $D((\hat{\mathcal{X}}_i, \hat{\mathcal{Y}}_i), (\hat{x}_i, \hat{y}_i))$ to $\tilde{\mathcal{D}}$
- 10 **end**
- 11 **end**
- 12 $\pi_i^{appr}(r_i^{in}) = \text{mean}(\tilde{\mathcal{D}})$
- 13 **end**
- 14 $r_i^{in} = \arg \max \pi_i^{appr}(r_i^{in})$

his strategy f_i^{in} is to randomize the query among four points with a fixed obfuscation distance r_i^{in} away at the four sides of his real location (x_i, y_i) :

$$\begin{aligned} & f_i^{in}((x'_i, y'_i) | (x_i, y_i), (\mathbf{x}'_{i-1}, \mathbf{y}'_{i-1})) \\ & = \begin{cases} \frac{1}{4}, & \text{if } (x'_i, y'_i) = (|x_i - r_i^{in}|, |y_i - r_i^{in}|), \\ \frac{1}{4}, & \text{if } (x'_i, y'_i) = (|x_i - r_i^{in}|, 1 - |y_i - 1 + r_i^{in}|), \\ \frac{1}{4}, & \text{if } (x'_i, y'_i) = (1 - |x_i - 1 + r_i^{in}|, |y_i - r_i^{in}|), \\ \frac{1}{4}, & \text{if } (x'_i, y'_i) = (1 - |x_i - 1 + r_i^{in}|, 1 - |y_i - 1 + r_i^{in}|). \end{cases} \end{aligned}$$

Otherwise if $(x_i, y_i) \in \mathcal{X}_i^{out}(\mathbf{x}'_{i-1}, \mathbf{y}'_{i-1}) = \mathcal{M} \setminus \mathcal{X}_i^{in}$,

$$\begin{aligned} & f_i^{out}((x'_i, y'_i) | (x_i, y_i), (\mathbf{x}'_{i-1}, \mathbf{y}'_{i-1})) \\ & = \begin{cases} \frac{1}{4}, & \text{if } (x'_i, y'_i) = (|x_i - r_i^{out}|, |y_i - r_i^{out}|), \\ \frac{1}{4}, & \text{if } (x'_i, y'_i) = (|x_i - r_i^{out}|, 1 - |y_i - 1 + r_i^{out}|), \\ \frac{1}{4}, & \text{if } (x'_i, y'_i) = (1 - |x_i - 1 + r_i^{out}|, |y_i - r_i^{out}|), \\ \frac{1}{4}, & \text{if } (x'_i, y'_i) = (1 - |x_i - 1 + r_i^{out}|, 1 - |y_i - 1 + r_i^{out}|). \end{cases} \end{aligned}$$

In the x - and y -domain, we can consider the obfuscated query $f_i((x'_i, y'_i) | (x_i, y_i), (\mathbf{x}'_{i-1}, \mathbf{y}'_{i-1}))$ separately for x_i and y_i . Given the reduction from 2D to 1D, we can apply Definition 3 for $f_i(x'_i | x_i, \mathbf{x}'_{i-1}, Q_i)$ and $f_i(y'_i | y_i, \mathbf{y}'_{i-1})$ in the 1D domain. Then the adversary's inference is consisted of $\hat{x}_i(x'_i | \mathbf{x}'_{i-1})$ and $\hat{y}_i(y'_i | \mathbf{y}'_{i-1})$ the same as in 1D. Therefore, user i 's expected privacy gain in (7) can be given in 2D as

$$\begin{aligned} \pi_i^{appr} &= \int_{(x'_i, y'_i) \in \mathcal{M}} \Pr((x'_i, y'_i) | (\mathbf{x}'_{i-1}, \mathbf{y}'_{i-1})) \\ & \min_{(\hat{x}_i, \hat{y}_i) \in \mathcal{M}} \int_{(x_i, y_i) \in \mathcal{M}} \Pr((x_i, y_i) | (x'_i, y'_i), (\mathbf{x}'_{i-1}, \mathbf{y}'_{i-1})) \\ & \quad D((\hat{x}_i, \hat{y}_i), (x_i, y_i)) dx_i dy_i dx'_i dy'_i, \end{aligned} \quad (19)$$

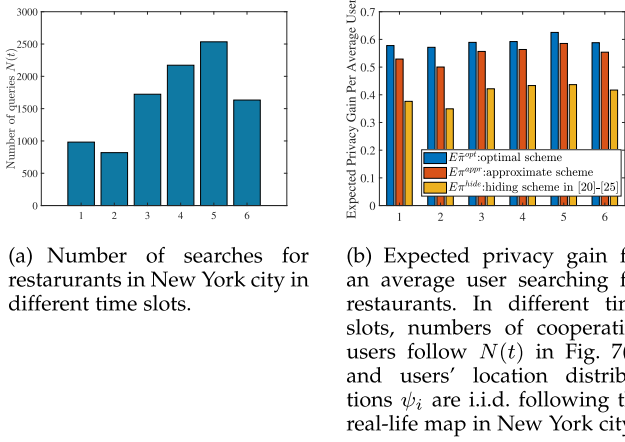


Fig. 7. Performance evaluation of our (four-point) approximate obfuscated query scheme ($\mathbb{E}\pi^{appr}$) in 2D from Algorithm 2 with our optimal strategy ($\mathbb{E}\pi^{opt}$) and the hiding scheme in the literature [20], [21], [22], [23], [24], [25]. Here, any user's service flexibility degree Q_i follows the i.i.d. truncated normal distribution with mean $\mu = 0.1$ and standard deviation $\sigma = 0.1$.

where $(\hat{x}_i, \hat{y}_i)$ is the adversary's inference attack similarly from (4). Similarly to Proposition 4, we can rewrite the user's privacy gain under the four-point obfuscation.

Proposition 9. By using a four-point approximate obfuscated query strategy in 2D in Definition 4, user i 's expected privacy gain in (19) can be rewritten by

$$\begin{aligned} \pi_i^{appr}(r_i^{in}, r_i^{out}, (\mathbf{x}'_{i-1}, \mathbf{y}'_{i-1})) \\ = \int_{(\hat{x}_i, \hat{y}_i) \in \mathcal{M}} \frac{1}{\sum_{x'_i} |\hat{\mathcal{X}}_i| \times \sum_{y'_i} |\hat{\mathcal{Y}}_i|} \sum_{x_i \in \hat{\mathcal{X}}_i} \sum_{y_i \in \hat{\mathcal{Y}}_i} \\ D((\hat{x}_i, \hat{y}_i), (x_i, y_i)) dx'_i dy'_i, \end{aligned} \quad (20)$$

where $\hat{\mathcal{X}}_i = \hat{\mathcal{X}}_i(r_i^{in}, r_i^{out}, \mathbf{x}'_{i-1}, \mathbf{y}'_{i-1})$ and $\hat{\mathcal{Y}}_i = \hat{\mathcal{Y}}_i(r_i^{in}, r_i^{out}, \mathbf{y}'_{i-1}, \mathbf{x}'_{i-1})$ can be obtained by solving equations from (15).

Algorithm 2 returns the optimal obfuscation distance r_i^{in} for Definition 3 with computational complexity $\mathcal{O}(\frac{1}{\epsilon^3})$.

Compared with Algorithm 1 in 1D, Algorithm 2 iterates similarly for the adversary's inference attack in one more dimension. To provide more practical performance evaluation of our approach, we use the datasets of real-world map [36] and search interests for the term 'restaurant' from Google Trends [37] in New York city for different time slots in a day. Under this general 2D ground plane scenario, users' locations are no longer uniformly distributed, and thus we next refine the adversary's inference procedure in Algorithm 2. As the real location x_i is not uniformly distributed, the candidates in sets $\hat{\mathcal{X}}_i(x'_i)$ and $\hat{\mathcal{Y}}_i(y'_i)$ for the real location are no longer inferred with equal probabilities. Therefore, we modify Step 5 in Algorithm 2 as

$$\text{Compute } \hat{x}_i(x'_i) = \sum_{x \in \hat{\mathcal{X}}_i(x'_i)} \frac{\psi_i^X(x) \cdot x}{\sum_{x \in \hat{\mathcal{X}}_i(x'_i)} \psi_i^X(x)},$$

where $\psi_i^X(x) = \sum_{y \in [0,1]} \psi_i(x, y)$ is user i 's general location distribution in the x -domain of the 2D plane. Step 8 in the y -domain can be modified similarly. For the user i 's expected

privacy gain in Steps 9 and 12, we modify the calculation as

$$\pi_i^{appr}(r_i^{in}) = \sum_{x \in \hat{\mathcal{X}}_i(x'_i)} \sum_{y \in \hat{\mathcal{Y}}_i(y'_i)} \frac{\psi_i(x, y) D((x, y), (\hat{x}_i, \hat{y}_i))}{\sum_{x \in \hat{\mathcal{X}}_i(x'_i)} \sum_{y \in \hat{\mathcal{Y}}_i(y'_i)} \psi_i(x, y)}.$$

Based on the dataset on Google Trends [37], Fig. 7(a) tells the number of searches for one kind of PoI (restaurant). Following the real-world population map in New York city [36], we further generate an i.i.d. location distribution ψ_i for all cooperative users. Fig. 7(b) then gives the comparison of $\mathbb{E}\pi^{opt}$, $\mathbb{E}\pi^{appr}$ and $\mathbb{E}\pi^{hide}$. Our approximate obfuscated query scheme is obviously better than the hiding scheme in the literature, and is close to optimal solution which suffers from high computational complexity.

VIII. CONCLUSION

To address potential data leakage from LBS platforms as well as peers, we study on multi-user privacy preservation problem under a cooperation scheme. We propose to leverage LBS users' service flexibility with one single query for a better privacy gain. Depending on whether a user benefits from the crowdsourced cache, he takes different query strategies with obfuscation in a distributed way and contribute to the cache to benefit latter users. Differently from the existing literature, we recommend users to query with obfuscated locations to the LBS platform even if they find useful PoI information in the cache, to jointly increase confusion to the adversary and store more PoI data. To relax the complexity, we simplify the query scheme to a two-point randomized query in 1D and a four-point randomized query in 2D. Both of them show guaranteed performance compared with the optimal scheme and significant privacy improvement compared with hiding schemes in the literature. From the perspective of the crowdsourced system, we also study the optimal query sequence to maximize the total expected privacy gain.

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